

Iris Structure

The iris is the most visible portion of the uveal tract, which is the middle compartment of the eye. The iris is made up of blood vessels and connective tissue, in addition to melanocytes and other pigmented cells that are responsible for its distinctive color.

The iris contains muscles that control its movement and allow for changes in the size of the pupil, controlling the amount of light that enters the eye and that ultimately reaches the retina. Though its function is easily observed and obvious to any who look at it under varying lighting conditions, it is the appearance of the iris—specifically, its color—that is most striking and apparent. The structure itself takes its name from Iris, the Greek goddess of the rainbow and messenger of the gods.

The iris consists of two layers of different embryological origin. The **anterior** border of the iris consists of the stroma, a loose and interrupted layer of connective tissue. It is composed of **melanocytes** and nonpigmented cells, as well as other types of cells and tissues. Melanocytes contain melanin, a brown or black pigment. The overall structure of the stroma is similar in irises of all colors. The iris pigment epithelium forms the densely pigmented **posterior** layer of the iris. It consists of two layers of tightly fused, pigmented cells.

Differences in iris color depends on the amount of pigmentation in the deep stroma, especially the anterior border layer, and on the density of the stroma, both of which influence how much light, and what wavelengths, are absorbed and reflected. As with other objects, the color we see is the result of reflected light. The stroma of brown irises is densely pigmented with melanin and absorbs much of the light that enters it. In many human populations, brown is the only eye color. Blue irises have lightly pigmented stroma, and light of longer wavelengths (red to yellow) readily penetrates the iris and is absorbed, while some light of shorter wavelength (blue) is reflected back and scattered by the iris stroma; hence the blue color.

The inability to make melanin, as in albinism, leaves the iris without any pigment. The iris appears pink from the color of the blood flowing through it. Albinism is a recessive condition, requiring two defective **alleles** for melanin production, one inherited from each parent. Albinism also prevents pigment production in the hair and skin.

Genetics of Eye Color

Differences in iris color have been attributed to such causes as the temperature of the brain and eyes. Some people have stressed differences between dark-eyed and light-eyed populations and have suggested that eye color is related to general traits such as temperament or intellect. But, toward the middle of the nineteenth century, it had become clear that iris color was due to iris pigment, that this pigment developed soon after birth, and that the final quantity and distribution of the pigment was a hereditary trait.

Originally, iris color was thought to be a simple trait—one governed by a single gene with multiple forms, or alleles, corresponding to each color. In this scheme, blue was thought to be recessive, requiring two copies of

anterior front

melanocytes pigmented cells

posterior rear

alleles particular forms of genes

the blue allele in order to be displayed. Therefore, two blue-eyed parents could have only blue-eyed children, since each parent had only blue alleles. However, repeated observation of brown-eyed offspring from two blue-eyed parents showed this view to be wrong. Iris color is likely to be a polygenic trait—one governed by at least two genes and possibly more.

Brown versus blue eye color is believed to be controlled by two genes on chromosome 15, called *BEY1* and *BEY2*. Green versus blue eye color is believed to be controlled by a gene on chromosome 15, called *GEY*. In this system, blue is believed to be recessive to both brown and green. The protein products of these genes are unknown, however, as is the number of alleles possible for each. Furthermore, these three do not fully explain inheritance of all eye colors. More genes, which likely modify the action of these three, are probably involved.

Traditionally, iris color was felt to be stable throughout adulthood. However, iris color may change in response to disease. For example, there is a gradual unilateral (one-sided) loss of pigmentation in Horner's syndrome and in Fuchs' heterochromic iridocyclitis. There is also evidence for pigment loss in the iris as a result of aging, and changes in iris color may also occur spontaneously in normal people after adolescence. In addition, some commonly used drugs such as latanoprost (which lowers intraocular pressure) have caused hyperpigmentation in some irises. SEE ALSO COMPLEX TRAITS; INHERITANCE PATTERNS.

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Fertilization

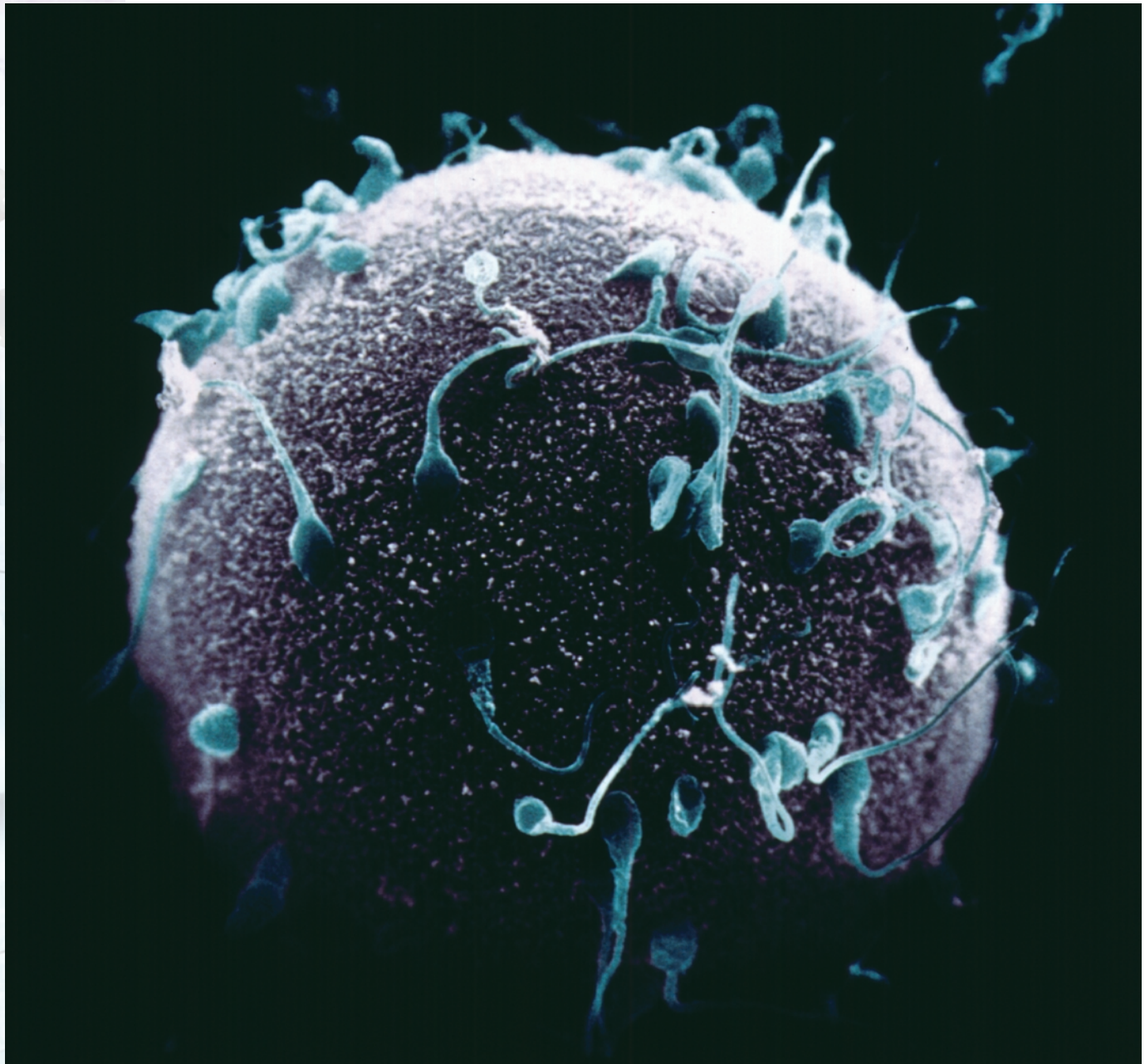
Fertilization is the fusion of a female's egg cell (oocyte) and a male's sperm cell (spermatozoa) to form the first cell of a new and unique being. While on the surface this sounds like a simple process, there are many factors that make this possible.

Gametes

Gametes are unique from all other cells. Typically, each cell in the human body contains twenty-three pairs of chromosomes (for a total of forty-six). Mature egg and sperm cells contain only one copy of each chromosome (for a total of twenty-three). At fertilization the fusion of the two gametes will create a cell with the appropriate twenty-three pairs of chromosomes

gametes reproductive cells, such as sperm or egg





A scanning electron microscope (SEM) image captures many sperm attempting to penetrate an egg during the process of fertilization. If one of these sperm is to enter the egg it must create a way in; its "head" releases enzymes to break down the egg's outer layer, the "zona pellucida."

(forty-six individual chromosomes) necessary for human development. In this way, one chromosome of each pair will originate from each parent, making the new individual unique from any other person that came before. The specialized process by which the genetic material is shuffled and the chromosome number is cut in half, from forty-six to twenty-three chromosomes, is called **meiosis**.

Ovulation and Ejaculation

Sperm and egg cells are not only different from other cells, but are different from each other. A female is born with all the eggs she will ever have. At birth, the chromosomes of these eggs have only completed the beginning of meiosis (meiosis I) and will remain dormant (inactive) until the onset

of menstrual cycles in puberty. Specific hormones produced during the menstrual cycle around day thirteen or fourteen trigger the continuation of meiosis in one egg each month. Meiosis is suspended for the second time in the middle of meiosis II around three hours prior to ovulation, and does not resume unless fertilization occurs.

During ovulation this egg, enclosed in two layers of protective material, is released from the ovary. The outer layer, the cumulus oophorus, is comprised of cells called cumulus cells, and the inner layer, the zona pellucida, is comprised of a jellylike coating made of protein and sugar. Once released from the ovary, the egg is swept into the fallopian tube. It is receptive to fertilization for only about one day. The sperm must reach the egg during this time, usually in the fallopian tubes, or fertilization will be impossible.

During sexual intercourse, millions of sperm are deposited into the vagina. They travel through the cervix and uterus to the fallopian tubes. Sperm can live within a woman's body for up to three days. Each sperm contains three distinct parts, the head, mid-piece, and tail. Each of these parts has a distinct purpose. The head of the sperm is composed of the nucleus (containing the chromosomes), an acrosome cap (containing **enzymes** crucial in fertilization), and an outer membrane. The mid-piece contains energy-producing mitochondria, and the tail is the mechanism for movement. Despite the ingenious design, fewer than 1 percent of the sperm released in an ejaculation ever make it to the egg. Factors inhibiting the success of sperm include abnormal formation and premature death from exposure to acidic vaginal secretions. Sperm can also be blocked by excess mucus covering the cervix, or they may travel to the fallopian tube that does not contain the egg. Fortunately, only one sperm is required to fertilize the egg.

In order to fertilize an egg, sperm must undergo the poorly understood process of capacitation. Capacitation involves changes to the acrosome, triggered by the cervical mucus, to prepare it to release the enzymes necessary to break through the zona pellucida. Upon reaching the surface of the zona pellucida, the sperm releases enzymes to break through. Once through the zona pellucida, the head of the sperm fuses to the egg's membrane, the tail of the sperm stops moving, and the egg engulfs the contents of the sperm.

It is crucial that only one sperm enters the egg. If an extra sperm passes through, a lethal condition known as polyspermy (many sperm) will occur. On the rare occasion this occurs, the fetus will be miscarried as a result of the extra set of chromosomes. To prevent this in most instances, a substance is released from the egg that changes the zona pellucida once it has been penetrated, blocking entry of any other sperm. Sperm penetration triggers the completion of the second meiotic division in the egg. With this division, the chromosomes of the sperm and egg come together in their own nucleus. The cell now officially becomes a **zygote**, the first cell of a new individual.

meiosis cell division that forms eggs or sperm

enzymes proteins that control a reaction in a cell

zygote fertilized egg

Variations

In some instances, an ovary releases more than one egg at one time, or both ovaries release an egg simultaneously. Each egg has the potential to be fertilized, resulting in multiple pregnancies. Since each conception originates



from a separate egg and sperm, individuals created in this way are as different as those conceived as separate births. These are referred to as fraternal twins. In rare instances, cells from the same embryo separate to create distinct embryos that are genetically identical and are referred to as monozygotic, or identical, twins.

The exact details of fertilization vary from animal to animal. Fertilization does not always take place inside an animal. For example, sea urchins, spiny animals attached to rocks on the ocean floor, release their eggs and sperm directly into the water. Large numbers of each (millions of eggs and billions of sperm) are necessary to ensure that enough eggs will be fertilized to maintain the population. Many other ocean creatures also release egg and sperm cells into the water. However, the eggs are fertilized only by sperm of the same species because of unique proteins on the surface of the egg. As in humans, fertilization immediately triggers a change in the surface of the egg, protecting it from penetration by other sperm, even sperm of the same species.

Reproductive technology has introduced further variations in how eggs may become fertilized, permitting the process to occur outside the fallopian tubes. One of the most common is **in vitro** fertilization, in which eggs and artificially capacitated sperm are combined in a glass dish and the dividing embryos are later transplanted into the uterus. **SEE ALSO** MEIOSIS; REPRODUCTIVE TECHNOLOGY; TWINS.

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Forensics See *DNA Profiling*

Founder Effect

The term "founder effect" refers to the observation that when a small group of individuals breaks off from a larger population and establishes a new population, chance plays a large role in determining which **alleles** are represented in the new population. The particular alleles may not be representative of the larger population. As the new population grows, the allele frequencies will usually continue to reflect the original small group.

Genetic Characteristics of Founder Populations

Because the founder population is small, **genetic drift** can play an important role in determining the genetic makeup of subsequent generations, and allele frequencies may fluctuate. For example, consider an extreme situation where a new population is founded by just two individuals, a male and a female, perhaps because they are stranded on an island. Assume that the

in vitro "in glass"; in lab apparatus, rather than within a living organism

alleles particular forms of genes

genetic drift evolutionary mechanism, involving random change in gene frequencies



The Hutterite colony, to which these children belong, is a founder population with a high prevalence of asthma.

mother is **heterozygous** for a particular allele (Aa), while the father is **homozygous** (AA). If the couple has two children, there is a 25 percent chance that the mother will pass the A allele to both children.

If neither child inherits the allele, the a allele is effectively lost from the population. Even as they grow, many founder populations remain relatively genetically isolated, with little immigration into the population. Examples include founder populations that have remained isolated due to geographical location, such as Finland and Iceland, or due to religious customs, such as Amish and Hasidic Jewish groups.

Founder populations may have increased prevalence of certain genetic traits, including genetic disease. Disease alleles that happen to be present in the founders may be passed on to offspring, and, since the population is small, there may be a higher prevalence of the disease than in other, larger populations. Isolated founder populations, with little marriage outside of the populations, are especially likely to have a higher prevalence of recessive disorders, since parents are likely to share many genes, and there is an increased chance of inheriting two copies of a particular disease gene. Examples of rare genetic diseases that are prevalent in founder populations are Tay-Sachs disease in Ashkenazic Jewish populations and asthma in the Hutterian population.

Founder Populations Can Be Valuable for Genetic Studies

The same forces that lead to increased risk of disease also make founder populations particularly useful for identifying which genes are involved in genetic disease. Since the founder population is derived from a small number of individuals, it is likely that those individuals with a particular disease have a common genetic profile, rather than having multiple different disease mutations or susceptibility alleles. This genetic homogeneity is important, since genetic heterogeneity can make identification of any particular disease allele very difficult.

heterozygous characterized by possession of two different forms (alleles) of a particular gene

homozygous containing two identical copies of a particular gene

In the nineteenth century, the rate of deafness on Martha's Vineyard (an island off the coast of Massachusetts) was thirty times that of the mainland population. Most deaf islanders were descendants of a small handful of English families who settled there around 1700.

locus site on a chromosome (plural, loci)

haplotype set of alleles or markers on a short chromosome segment

Linkage disequilibrium mapping is a powerful method for fine-mapping disease genes in founder populations. Linkage disequilibrium refers to the physical association between harmless but traceable marker alleles and a disease allele on a chromosome. The close proximity of the markers can help pinpoint the disease **locus**. Founder populations are particularly useful for linkage disequilibrium mapping since regions in linkage disequilibrium often span greater chromosomal distances than in general populations; that is, the disease gene will often be found with a larger set of common markers in a founder population than in a larger, more diverse population. This is expected because in founder populations, all chromosomes carrying a specific disease allele may be descended from a single ancestral chromosome, thus the disease allele will be in linkage disequilibrium with alleles at nearby markers. In a larger, more diverse population, the disease allele may have arisen on several different chromosomes, therefore the linkage disequilibrium, even for very close markers, may not be as great.

One example of linkage disequilibrium mapping in founder populations is the identification of a region containing the diastrophic dysplasia gene in eighteen families from Finland. This condition causes bent or abnormal bone growth. The region to which the disease gene was localized was narrowed substantially because scientists were able to take advantage of the extensive linkage disequilibrium around this gene in the affected individuals, all of whom shared a series of alleles surrounding the disease gene.

A related method for mapping disease genes that is well-suited for founder populations is **haplotype** analysis. A haplotype is defined as the set of alleles that are inherited as a group from one parent. A haplotype forms an identifiable pattern that can be used to track inheritance of all the genes within it. There are only a small number of haplotypes among the founders. Recombination tends to break up haplotypes over time, with the alleles that are closest together remaining together the longest.

A haplotype that is constantly inherited with a disease can be analyzed to narrow the region in which the gene should be sought. This means that researchers can look for shared regions or segments of chromosomes among affected individuals to help identify the location of a disease gene. For example, genetic researchers were able to demonstrate that the majority of cases of idiopathic torsion dystonia (a neurological disease) in Ashkenazic Jews were due to a single mutation from a common ancestor, because the affected individuals shared common alleles (a consistent haplotype) on either side of the mutation.

Another advantage of genetic studies in founder populations is that good clinical and genealogical recordkeeping is often available. Many genetic studies have been successful in Finland because of the population history of this region. For instance, the current Finnish population is believed to have come from a small group of individuals who settled in the southwest part of the country about 2,000 years ago. Since the initial immigration, the population has continued to be relatively isolated, with little migration into it.

Genealogical records are available through church parishes and often go back six to twelve generations, allowing scientists to develop accurate and detailed family histories linking individuals together. Despite these advantages, for common diseases such as asthma, scientists must consider that

genes that cause asthma in Hutterites may or may not be relevant to other groups with asthma. Thus the scientist must weigh the advantages of performing genetic studies in small, historically isolated populations with the potential disadvantage of being unable to eventually generalize the studies' results. SEE ALSO GENETIC DRIFT; HARDY-WEINBERG EQUILIBRIUM; INBREEDING; LINKAGE AND RECOMBINATION; MAPPING; POPULATION BOTTLENECK; POPULATION GENETICS; TAY-SACHS DISEASE.

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Fragile X Syndrome

Fragile X syndrome is one of the most common causes of inherited mental retardation. Individuals with fragile X syndrome can exhibit moderate to severe mental retardation. Additional characteristics may include autistic-like behavior, hyperactivity, mitral valve prolapse (a heart valve defect), a large head circumference, a long face with a prominent forehead and jaw, protruding ears, flat feet, hyper-extensive joints ("double-jointedness"), and, in males, enlarged testicles. Fragile X syndrome is not restricted to any ethnic group. It was the first of the so-called triplet repeat diseases to be discovered, and study of it has led to a growing understanding of DNA instability and its role in disease.

Discovery of the Syndrome

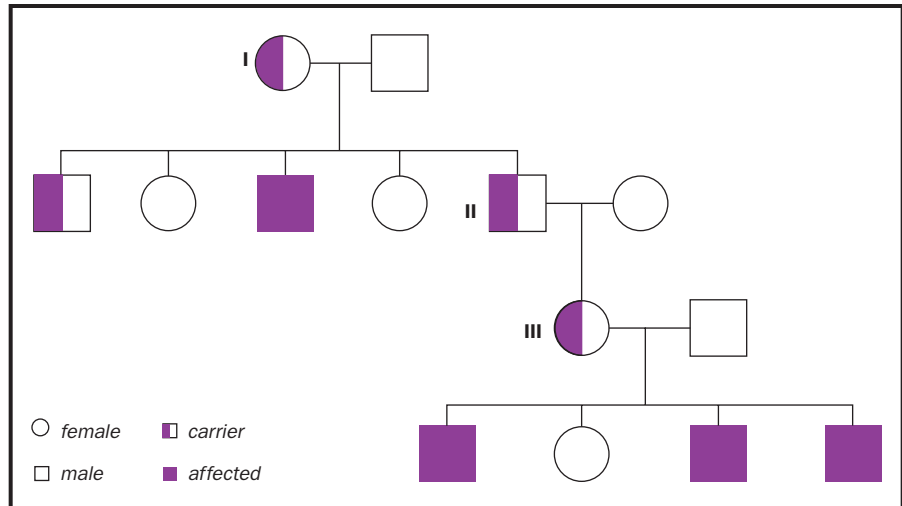
The first family with fragile X syndrome was described by J. Purdon Martin and Julia Bell in 1943. This family had eleven severely retarded males, and the inheritance pattern of the mental retardation appeared to be X-linked. X-linked traits are inherited on the X chromosome and are more common in males, who have only one X chromosome, than in females, who have two.

In 1969, in a different family, Herbert Lubs observed a constriction near the end of the long, or q, arm of the X chromosome in four mentally retarded males and two of their mentally normal female relatives. This constriction made the X chromosome appear to be broken. Hence the name "fragile X."

For years, little attention was paid to Lubs's finding. Renewed interest in the observation emerged in the late 1970s, when additional families were identified with mental retardation and the same chromosome abnormality, or fragile site. Moreover, in 1977 Grant Sutherland discovered that the ability to detect this fragile site was dependent on the chemicals used to study patients' chromosomes. Sutherland's crucial observation helped develop the



The Sherman Paradox. The daughter (III) of an unaffected male carrier (II) is more likely to have affected offspring than the mother (I) of the male carrier is.



first diagnostic test for fragile X syndrome. Using this knowledge, investigators reexamined chromosomes from the original fragile X family described in 1943 and demonstrated that, indeed, affected individuals in this family carried the characteristic fragile site.

Puzzling Inheritance Pattern

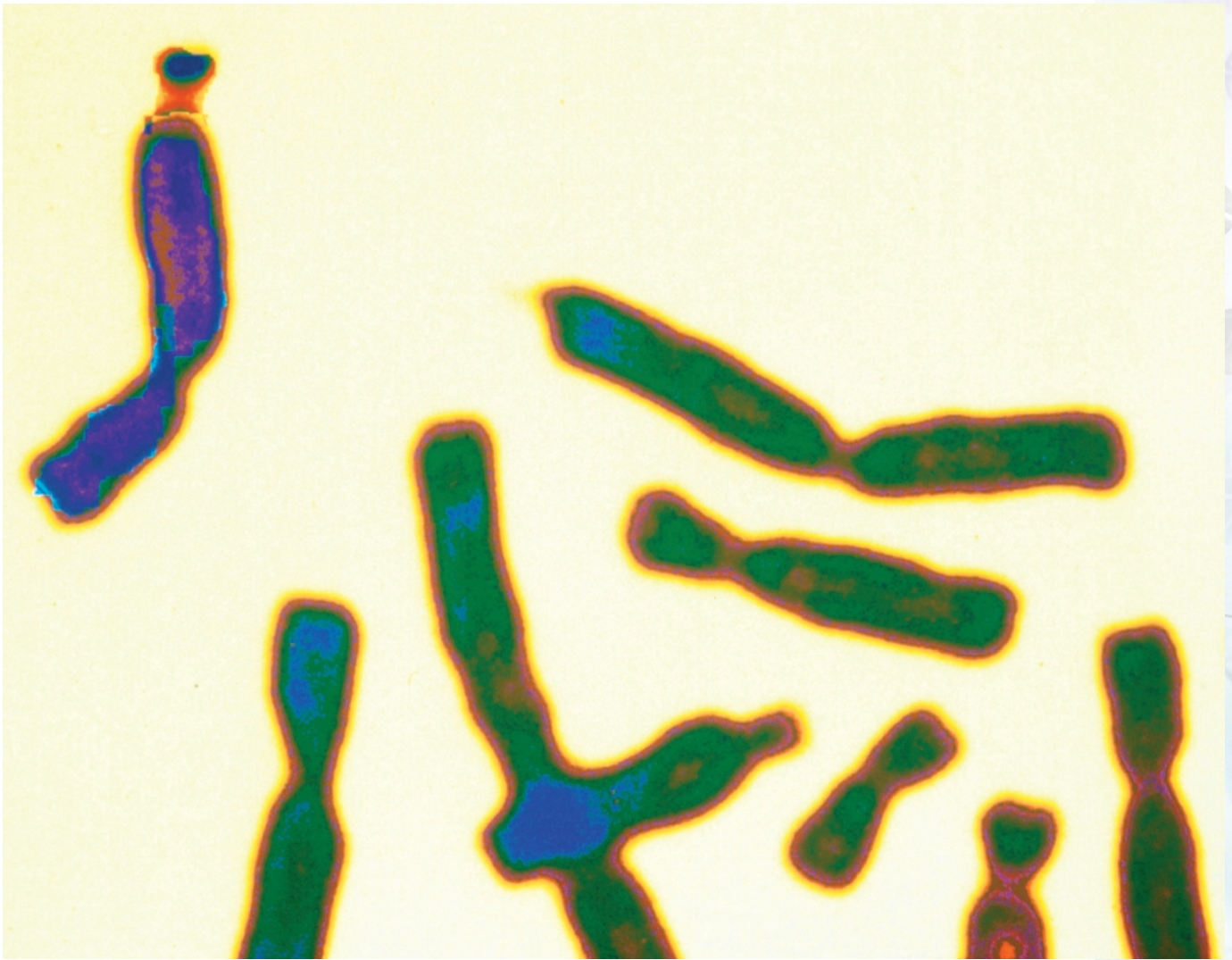
While the location of this fragile site established that fragile X syndrome was indeed X-linked, inheritance of this disorder was clearly not typical of other X-linked disorders. At first, it was believed that fragile X syndrome was an X-linked recessive genetic disorder. However, there were many observations inconsistent with this inheritance pattern.

If the disorder was truly inherited in an X-linked recessive manner, **heterozygote** carrier women would not display any characteristics of the syndrome, and all carrier males would. But there were reports of affected females, and of males who carried the fragile site but were unaffected. It was particularly difficult to reconcile that some male carriers could be so severely affected while others were completely unaffected.

Because of these puzzling observations, in 1985 Stephanie Sherman and her colleagues studied the inheritance pattern of fragile X syndrome more closely. They demonstrated that the risk of expressing mental retardation was dependent on the individual's position in the pedigree, with risk increasing in later generations. The daughter of an unaffected male carrier was more likely to have affected offspring than the mother of the unaffected male carrier was: something had changed on the X chromosome over the two generations. This observation became known as the "Sherman paradox" and was crucial to understanding the genetic mutation that causes fragile X syndrome.

To explain the unusual inheritance pattern, Sherman, her colleagues, and several other scientists hypothesized that the alleged gene for fragile X syndrome was mutated in a two-step process. They proposed that the first mutation caused a "premutation" state that produced no clinical symptoms, and that a second mutation was required to convert the premutation to a "full mutation" form that was associated with the characteristic symptoms

heterozygote an individual whose genetic information contains two different forms (alleles) of a particular gene



of fragile X syndrome. Moreover, conversion from a premutation to a full mutation was proposed to occur only when the premutation was transmitted from a carrier female.

An Expanding Gene

In 1991 an international team of scientists identified the gene and mutation that causes fragile X syndrome. They found that in families with fragile X syndrome, there is a piece of the *FMR1* gene, called a CGG repeat, which is abnormally expanded.

In the general population, the repeat length can range from about six to fifty-four copies of the CGG, and the repeat is stable, or is passed from parent to child without change. In fragile X families, the premutation form of the repeat contains between fifty and two hundred copies of the CGG repeat, and the repeat is unstable.

Premutation alleles can expand to full mutation **alleles** (with more than two hundred copies of the CGG repeat) by transmission of the premutation from a mother to her child. A woman's risk of having a child with the

The fragile X chromosome is stained with purple dye. The affected chromosome is described as "fragile" because of the extra constriction near the end of the long arm of the X chromosome. Mental disability presents in 80 percent of carrier males, and 50 percent of carrier females.

alleles particular forms of genes

full mutation correlates to her own repeat size. The larger her premutation, the more she risks having a child who carries the full mutation.

The CGG repeat is usually interrupted by a single AGG trinucleotide every ten CGG repeats, but this can vary from individual to individual. Because premutation alleles have fewer AGG interruptions compared with normal-size *FMR1* alleles, it is believed that the AGG interruptions are important for stability of the CGG repeat.

Individuals with a premutation do not express the clinical symptoms associated with fragile X syndrome, although it has been reported that premutation carrier females can experience premature ovarian failure. Individuals who carry the full mutation can express symptoms of fragile X syndrome because they are missing the protein produced by the *FMR1* gene. Males with a full mutation always exhibit some symptoms of the disorder. Due to X inactivation, females with a full mutation may or may not express symptoms.

Although there is currently no cure for fragile X syndrome, scientists are making great progress in understanding the biology of the disorder. In the mid- to late 1990s, Stephen Warren and colleagues determined that the *FMR1* gene product, named FMRP, is an RNA-binding protein that shuttles in and out of the nucleus and is involved in binding various messenger RNAs. Moreover, scientists successfully developed mice that lack the *FMR1* gene, which will greatly aid research. Symptoms of fragile X mice include learning disabilities, hyperactivity, and, in males, enlarged testicles. Prevailing hypotheses about FMRP suggest that this protein is involved in forming neural connections in the developing brain.

The identification of *FMR1* and the expanded CGG repeats was a landmark discovery in human genetics because it established a novel class of human genetic mutations, trinucleotide (or triplet) repeat expansions. Since the discovery of *FMR1* and the expanding CGG repeats, scientists have identified more than ten other human genetic disorders that are caused by expansions of trinucleotide repeats, including disorders such as Huntington's disease and myotonic muscular dystrophy. SEE ALSO INHERITANCE PATTERNS; INTELLIGENCE; MOSAICISM; TRIPLET REPEAT DISEASE; X CHROMOSOME.

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Fruit Fly: *Drosophila*

Drosophila melanogaster, a common fruit fly, was one of the first model organisms used in genetic research, and continues to be one of the most important. Thomas Hunt Morgan (1866–1945) developed *Drosophila* as a model system in 1909. Morgan, along with his students, Calvin Bridges, Alfred

Sturtevant, and Hermann Muller, made some of the most important discoveries in genetics through their work with *Drosophila*. Among these were the genetic explanation of sex linkage (the location of a gene on a sex chromosome); proof that genes are contained on chromosomes; and the demonstration that genes are arranged on a chromosome in a linear order with fixed, measurable distances between them, the principle that underlies genetic mapping.

Like other good model organisms, *Drosophila* is easy to rear in the laboratory. It has a short life cycle, lasting about two weeks, and produces many offspring. Each female can lay hundreds of eggs. These traits make it ideal for isolating mutants and carrying out many genetic crosses rapidly.

Mutants are the cornerstone of genetic analysis. To find a mutation one must be able to recognize an observable physical trait, or **phenotype**, such as a change in anatomical structure or behavior. At first glance, watching a tiny fruit fly landing on a rotting banana, one may be hard pressed to imagine that anyone could spot an anatomical variant, much less begin to study such a complex subject as behavior. Observed through a low-powered microscope, however, *Drosophila* is a sculptural masterpiece of bristles, segments, colors, and mosaic patterns. By studying *Drosophila* mutants, scientists have devised ways to genetically dissect the cellular bases of these phenotypes, as well as such startlingly complex behaviors as learning, memory, and even sleep.

A feature of a model organism that aids geneticists is a small genome size and a small number of chromosomes, since the less DNA there is to sort through, the easier it is to find genes. *Drosophila*'s genome, containing about 180 million base pairs, is approximately one-twentieth the size of the human genome. There are four pairs of chromosomes: the X and Y sex chromosomes, and **autosomes** 2, 3, and 4. The complete nucleotide sequence of the gene-rich portion of the genome was determined in 2000. The genome is estimated to encode approximately 13,000 genes.

Drosophila molecular geneticists make wide use of **transposons**. These are short segments of DNA that, when injected into a cell, can insert themselves into the chromosomal DNA at random positions. Using recombinant DNA methods, a researcher can splice any gene into a transposon, which can then serve as a **vector** for introducing the gene into a fly.

Alternatively, transposon insertion can be used to cause mutations in genes. While much of the chromosomal DNA consists of sequences that code for non-protein elements, such as introns and “spacer” sequences between genes, a transposon may become inserted directly into a protein-coding sequence. This usually alters the amino acid sequence of the protein encoded by a gene, rendering the gene product dysfunctional. Even without knowing which gene was mutated, or where in the genome it is located, a researcher can make use of the transposon insertion as a “molecular tag” to rapidly identify the gene. Since the sequence of the transposon is known, a DNA probe can be designed to detect it (and therefore find the gene which it has mutated) by molecular hybridization methods.

An unusual phenomenon of the chromosomes in certain of *Drosophila*'s tissues provides a powerful tool for determining the positions of individual genes. The chromosomes in the fruit fly's salivary gland cells replicate

phenotype observable characteristics of an organism

autosomes chromosomes that are not sex determining (not X or Y)

transposons genetic elements that move within the genome

vector carrier



Because of their ability to produce large amounts of offspring in a short amount of time, fruit flies were ideal specimens for early genetic experiments.



metaphase stage in mitosis at which chromosomes are aligned along the cell equator

several hundred times without separating from each other by mitosis and cell division. Instead, the newly replicated DNA strands line up parallel to one another to form a tight bundle, called polytene chromosomes. Polytene chromosomes are less condensed than normal **metaphase** chromosomes, and can be around 2 millimeters long, large enough to easily be examined in detail under a low-power microscope. When stained with certain dyes, polytene chromosomes display characteristic banding patterns along their length. *Drosophila* geneticists have made maps of the banding patterns and have learned to use them as landmarks to help them locate genes of interest. Polytene chromosomes make *Drosophila* an excellent organism for the sub-branch of genetics known as cytogenetics, which is genetic analysis through directly visualizing the chromosomes themselves.

One of the most important areas of research to come from studies on *Drosophila* is that of embryonic development. By analysis of mutants, developmental biologists have elucidated a complex and precise picture of how

genes orchestrate the development of a fertilized egg into an adult fly. Many of the genes exercising the master control over these processes encode transcription factors, which are proteins that regulate when and where particular genes are transcribed to produce messenger RNA. The initial set of genes in the hierarchy acts in the **oocyte**, even before it is fertilized by a sperm. Their main function is to define the spatial polarity of the oocyte, determining which is the front and rear (posterior and anterior polarity), and which is the belly and back (dorsal and ventral polarity). These genes, in turn, activate genes that divide the embryo into segments and subsegments, which will eventually become the body segments of the adult animal. Later-acting genes, termed homeotic genes, act within each segment to define its identity, for example a wing or a leg. Remarkably, the genes and regulatory pathways involved in *Drosophila* development are highly conserved (that is, very similar genes and pathways are present), not only in other invertebrates, but also in mammals, including humans. SEE ALSO CHROMOSOME, EUKARYOTIC; DEVELOPMENT, GENETIC CONTROL OF; *IN SITU* HYBRIDIZATION; MODEL ORGANISMS; MORGAN, THOMAS HUNT; MULLER, HERMANN; MUTAGENESIS; MUTATION; TRANSCRIPTION FACTORS; TRANSPOSABLE GENETIC ELEMENTS.

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Gel Electrophoresis

Gel electrophoresis is a widely used technique for separating electrically charged molecules. It is a central technique in molecular biology and genetics laboratories, because it lets researchers separate and purify the nucleic acids DNA and RNA and proteins, so they can be studied individually. Gel electrophoresis is often followed by staining or blotting procedures used to identify the separated molecules.

Basic Procedure

In electrophoresis, an electric field is generated to separate charged molecules that are suspended in a matrix or gel support. Negatively charged molecules move toward the **anode**, on one side of the gel, and positively charged molecules move toward the **cathode**, on the other side. The gel itself is a porous matrix, or meshwork, often made of carbohydrate chains. Molecules are pulled through the open spaces in the gel, but they are slowed down by the meshwork based on their differing properties.

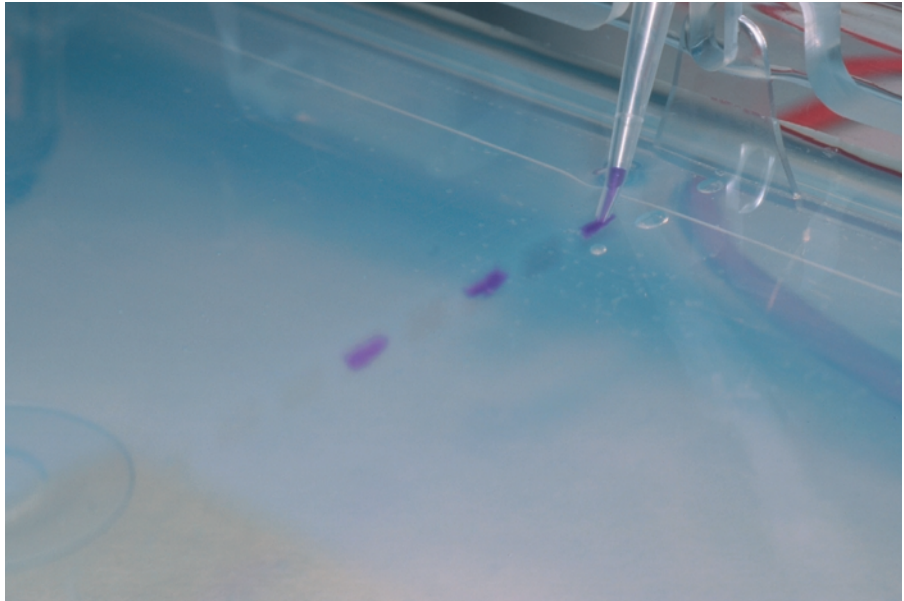
oocyte egg cell

G

anode positive pole

cathode negative pole

A solution containing DNA is added to the gel using a micropipette.



The parameters that determine the migration rate of these molecules through the meshwork are the strength of the electric field, the composition of the gel support or matrix, the composition of the liquid buffer solution the gel sits in, and the size, shape, charge, and chemical composition of the molecules being separated. Smaller molecules move faster than larger molecules, because they encounter less frictional drag in the gel. The size of the pores in the gel can be changed so this frictional drag is increased or decreased, allowing faster separation, or finer resolution.

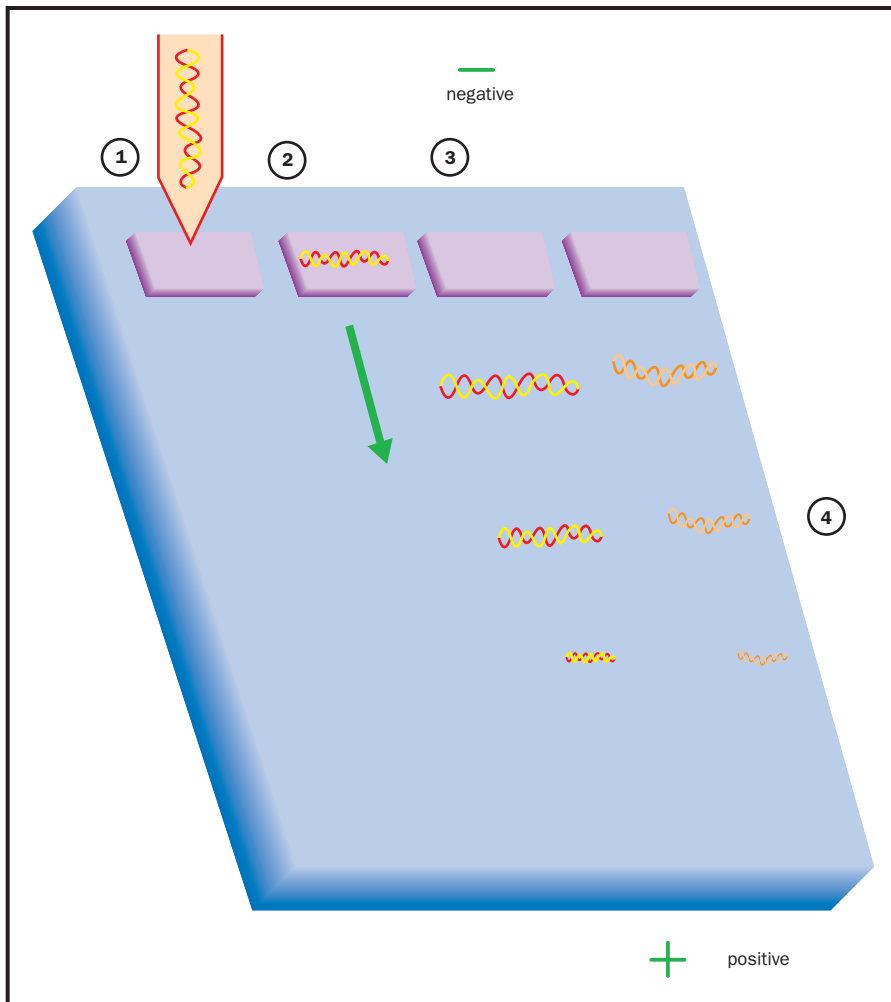
The electrophoretic technique can analyze and purify a variety of biomolecules, but is mainly used to separate nucleic acids and proteins. A basic consideration for choosing this technique is the composition of the sample to be separated—for example, does it contain nucleic acids (DNA or RNA), or is it composed of proteins, or carbohydrates? What are the sizes of the molecules to be separated? Another important consideration is the purpose of the separation—is it qualitative, where the technique is being used to evaluate the composition of the sample, or is it quantitative, in that the separated materials are to be collected for further analysis? Cellulose or starch is used as a support medium for low molecular-weight biomolecules such as amino acids and carbohydrates, whereas separation of proteins and nucleic acids are almost always done in gels made of a porous insoluble material such as agarose or acrylamide.

Separation of Proteins

PAGE. Proteins are usually separated using vertical polyacrylamide gel electrophoresis (PAGE), a process that separates them on the basis of their electric charge and their size. Proteins with a greater negative charge will be attracted more strongly and move faster toward the anode. The **charge density** on the proteins would cause smaller molecules to move more quickly through the gel's pores.

The size of a gel's pores can be changed depending on the size range of the proteins being separated. This is done by raising or lowering the con-

charge density ratio of net charge on the protein to its molecular mass



(1) DNA fragments are loaded into the gel. (2) Electric current is turned on, pulling the negatively charged DNA toward the positive electrode. (3) Smaller DNA fragments move faster through the gel. (4) The DNA is stained, revealing its position.

centration of acrylamide and bisacrylamide in the gel. Increasing the concentration results in more crosslinking between the two components, decreasing the pore size. Decreasing the concentration increases the pore size of the gel. Small proteins are separated better in a gel with large pores.

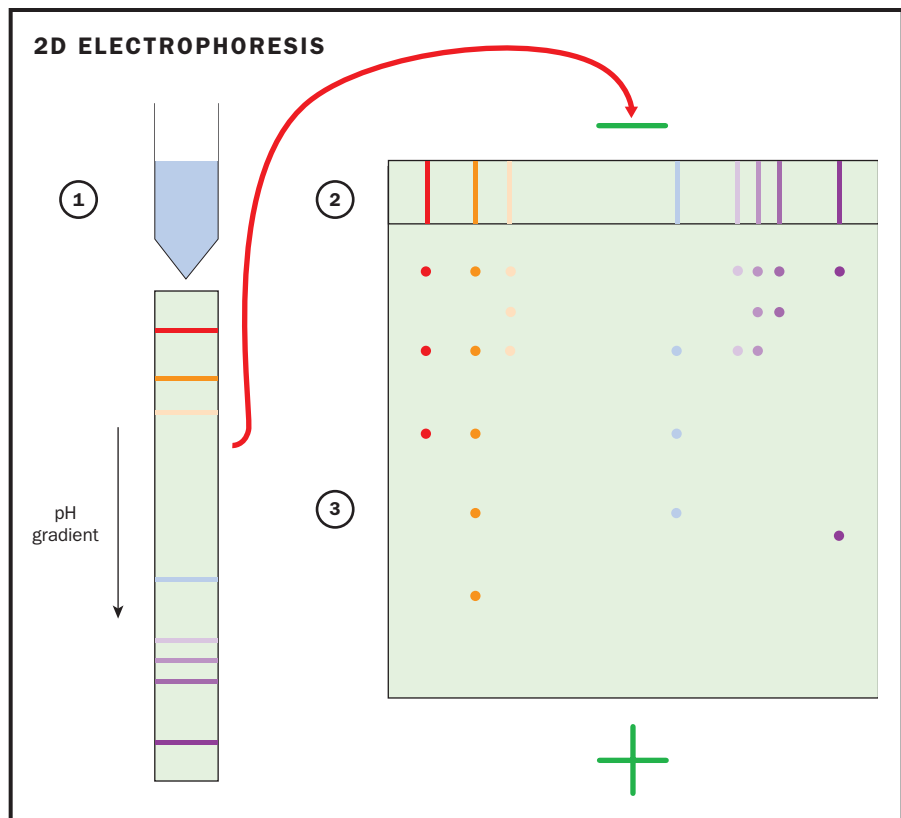
SDS-PAGE. Protein activity, such as for **enzymes**, can be determined once they are separated in the gel under conditions that do not **denature** the enzyme. Researchers can determine a purified protein's molecular weight by measuring how quickly the protein moves through a gel. The protein is first purified and denatured with heat and a reducing agent that disrupts disulfide bonding. It is then treated with an anionic detergent, sodium dodecyl sulfate (SDS), which disrupts the secondary, tertiary, and quaternary structure of the protein and coats it uniformly with negative charges. When run through a gel, the protein's migration rate is indirectly proportional to the logarithm of its molecular weight, so the smaller protein runs the fastest. The uniform negative charges ensure that the protein's migration rate is a function only of its molecular weight, not of whatever charges happen to be on it. Better resolution and separation can be obtained in an SDS-PAGE gel by first tightening the protein band before separating the proteins by size. This is accomplished by having the separating, or resolving, gel on the bottom and the larger pore gel, known as the stacking gel, on top. The

enzymes proteins that control a reaction in a cell

denature destroy the structure of



(1) A protein mixture is first separated by pH. (2) All proteins are made negative by the addition of a detergent, and are separated by size. (3) The final gel is stained to reveal the position of hundreds of unique proteins.



proteins enter the top gel, where they are maintained by in a tight zone between ions generated by the electric field. This is accomplished by having ions that run both slower and faster than the negatively charged proteins, so that ions sandwich the proteins between them, tightening the protein band. The proteins leave the stacking gel and enter the separating gel. In this gel the ions no longer sandwich the proteins because of a change in pH, and the pore size is smaller so that the proteins separate by size.

If the protein is run on a gel along with a ladder of proteins of known weight, then the molecular weight of the protein can be determined by comparing its migration rate to that of proteins whose molecular weights are known. This technique is known as “sodium dodecyl sulfate–polyacrylamide gel electrophoresis,” or SDS-PAGE.

Isoelectric Focusing. Researchers can use gels to determine a protein’s “isoelectric point,” or the pH at which the protein’s net charge is zero. Because pH changes the ionization state of several amino acid groups, the net charge on a protein is pH-dependent. By running proteins through a gel that has a pH gradient from one end to the other, this charge is gradually changed. At a certain pH, each protein’s net charge will become zero, and the protein will stop moving. This procedure is known as “isoelectric focusing.”

Two-Dimensional Electrophoresis. In “two-dimensional electrophoresis,” a mixture of proteins is first separated in an isoelectric focusing tube gel. This tube is then placed sideways on an SDS-PAGE gel. In this way, proteins are separated based on two parameters: size and isoelectric point. Com-

pared to techniques based on only one parameter, two-dimensional electrophoresis separates more proteins at once.

Two-dimensional electrophoresis is an important tool in **proteomics**. It can be used to separate large numbers of proteins that are isolated all at once after being expressed in response to a hormone, drug, or other stimulus. It can be combined with the use of DNA microarrays to allow a researcher to determine both what genes are expressed in response to a stimulus and what proteins are produced by these genes, which thereby determine an organism's physiological response to stimuli.

Separation of DNA and RNA

Nucleic acids come in a very wide range of sizes, from several dozen base pairs to many millions. No single technique can be used to separate them all. Instead, researchers analyze the nucleic acid molecules using the overlapping electrophoretic techniques of polyacrylamide, agarose, and pulse-field gel electrophoresis. Each technique places DNA or RNA molecules in an electric field. Because the nucleic acid fragments contain negatively charged phosphate groups along the backbone of the DNA molecule, they move toward the positively charged anode. As with proteins, the migration rate of nucleic acids through a gel depends on their conformation, the buffer composition, the concentration of the gel support, and the applied voltage.

Agarose Gels. The techniques discussed so far are good for separating proteins and small nucleic acid fragments from 5 to 500 **base pairs**. The small pores of the polyacrylamide gels, however, are not appropriate for larger DNA fragments or intact DNA molecules such as plasmids. Gels made of agarose, a natural seaweed product, are used to characterize nucleic acids that are 200 to 500,000 base pairs long.

Agarose gels, which can be purchased commercially, are prepared by dissolving purified agarose in warm electrophoresis buffer, cooling the solution to 50 °C (122 °F), and then pouring it into a mold, where it turns into a gel. Just as with polyacrylamide, the concentration of agarose in a gel determines the size of its pores. A comb placed in the gel before it sets produces the wells necessary for loading nucleic acid samples.

Nucleic acid fragments that are to be separated by size must be in “linearized” form. Plasmids, for example, must have their circular structure cut open using **restriction enzymes** before they are run on the gel. Otherwise, their rate of migration will depend on how supercoiled they are and whether they are nicked, instead of on their size. Nucleic acids that are separated in a gel can be seen with ethidium bromide or other stains.

Pulse-Field Gel Electrophoresis. The conventional agarose gel electrophoresis described above separates nucleic acid fragments smaller than 50,000 base pairs (50 kilobase pairs). Pulse-field gel electrophoresis separates huge pieces of DNA that are between 200 and 3,000 kilobase pairs long. In this technique the electric field is not held constant during the separation. Instead, its direction and strength are repeatedly changed, with the molecules reorienting themselves every time the current changes. The molecules then slither like a snake through the gel matrix, in a process known as “reptation,” with smaller fragments moving faster than larger ones. As the gel runs, it heats up and becomes more fluid, with the pulsing allowing

proteomics the study of the full range of proteins expressed by a living cell

nucleic acids DNA or RNA

base pairs two nucleotides (either DNA or RNA) linked by weak bonds

restriction enzymes enzymes that cut DNA at a particular sequence



the larger pieces to move more easily the longer the gel runs. Typically such gels are run overnight.

Once separated, large DNA pieces, such as complete genes, can be isolated for further experiments. They can be cloned into a bacterium, sequenced, or amplified by polymerase chain reaction. SEE ALSO **BLOTTING**; **CLONING GENES**; **DNA MICROARRAYS**; **POLYMERASE CHAIN REACTION**; **PROTEINS**; **PROTEOMICS**; **PURIFICATION OF DNA**; **SEQUENCING DNA**.

Linnea Fletcher

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Gene

Genes are functional units of DNA that contain the instructions for making proteins or RNA. Genes also act as units of heredity, transferring the same instructions from parent to offspring. The nature, structure, and regulation of genes has been a central topic of scientific research for more than 100 years.

History of the Gene and Structure of DNA

Genes were first defined as units of hereditary transmission. The name “gene” was coined by Wilhelm Johannsen in 1909, although the concept of a discrete unit governing inherited characteristics goes back at least to Gregor Mendel in 1861. The work of Thomas Hunt Morgan and his colleagues established that genes were located on chromosomes, and in the mid-1940s Oswald Avery demonstrated that genes were composed of DNA (deoxyribonucleic acid). Since that time, some types of viruses have been discovered that use ribonucleic acid (RNA) instead of DNA, but here we shall concentrate on DNA genes. The discovery of the structure of DNA in 1953 by James Watson and Francis Crick set the stage for the next fifty years of research into gene structure, function, and regulation.

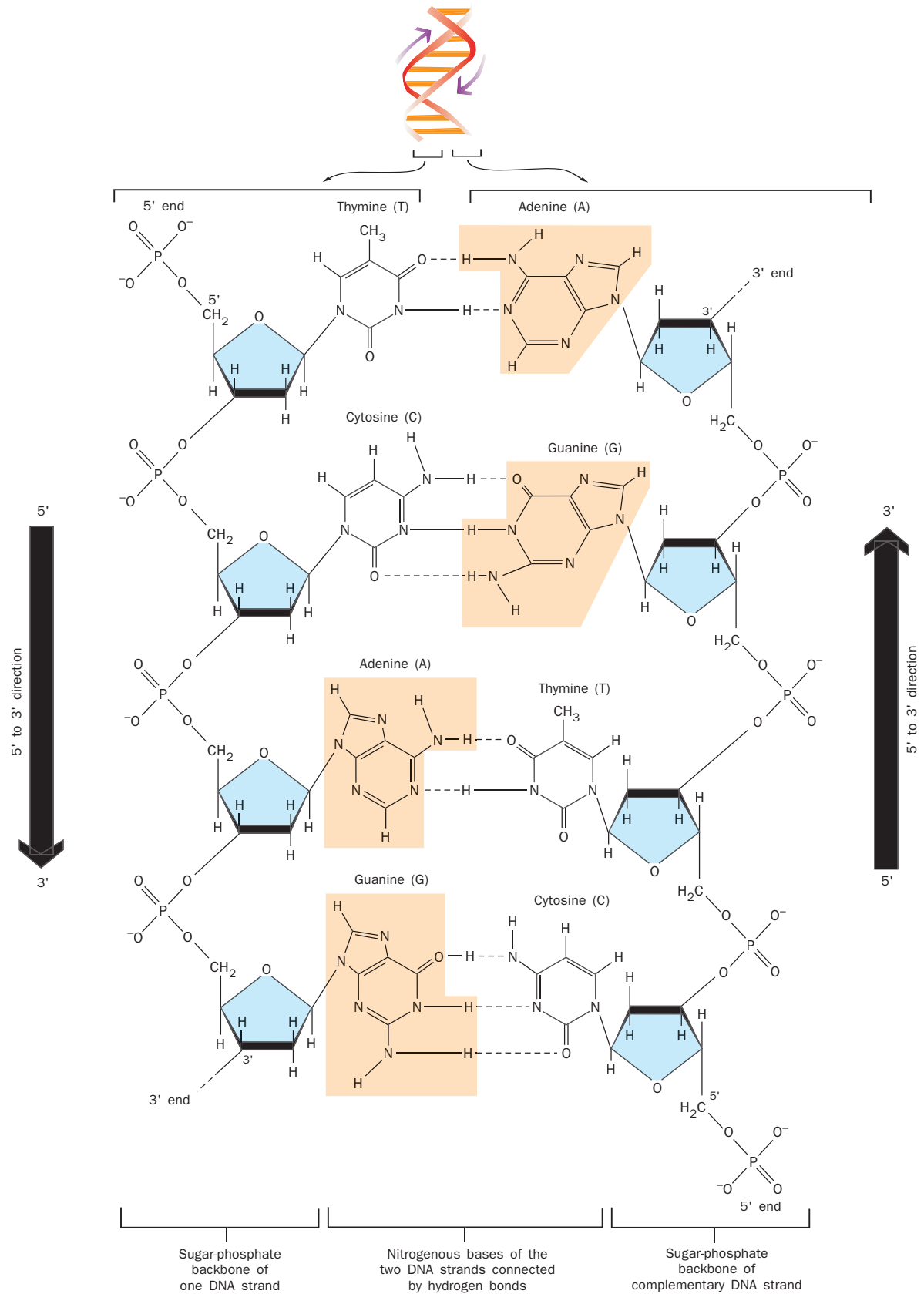
DNA is a linear molecule composed of subunits called **nucleotides**. Each nucleotide is made of a sugar and phosphate group, plus a chemical base, of which there are four types: adenine, thymine, guanine, and cytosine (A, T, G, C). Nucleotides are typically referred to by the name of their base. DNA exists as a pair of strands, wound around one another into a double helix, with the bases directed into the center. The structure and charges of the bases dictate that A on one strand can match only up with T on the other, and C only with G. This complementarity provides the basis for faithful replication of the entire DNA molecule.

Genes Code for Protein and RNA

While all genes are made of DNA, not all stretches of DNA act as genes. Indeed, in **eukaryotic** organisms, most of the DNA does not function as genes, meaning it is not the code for making proteins or RNA. Some DNA outside of genes has a structural role, some are remnants of old genes that

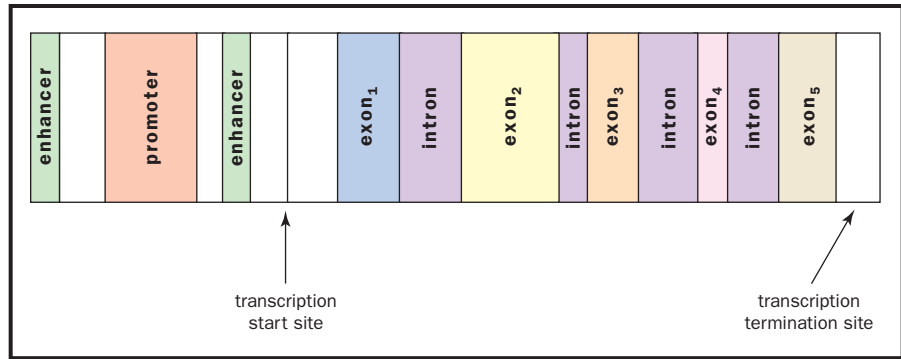
nucleotides the building blocks of RNA or DNA

eukaryotic describing an organism that has cells containing nuclei



DNA nucleotides pair up across the double helix.

Schematic structure of a gene. Only the exons (shaded) code for the final product, which may be protein or RNA.



now are functionless, and much of it appears to be “junk,” inserted and copied by viruslike sequences. Within a gene, usually only one side of the double helix actually codes for product; the other side is silent. Which side of the helix acts as code varies from gene to gene.

Almost all genes code for proteins. Proteins are strings of **amino acids**, and the sequence of nucleotides in the gene dictates the sequence of amino acids in the protein. Proteins perform almost all the functions in cells, and can be grouped into four major classes: they act as enzymes that control the rate of chemical reactions in the cell; they form structural components of organelles, membranes, and other cell components; they receive and transmit signals between and within cells; or they act as regulators of genes by latching onto DNA, thereby increasing or decreasing the rate at which the gene is used, or “expressed.”

Genes vary in length. The largest human gene is 2.5 million **base pairs** in length, and codes for the muscle protein named dystrophin, which is more than 3,500 amino acids long. Eukaryotic genes generally produce proteins of about 150 to 3,000 amino acids in length. Some genes are relatively small, as in **prokaryotes**, which produce proteins of 50 to 300 amino acids. Most eukaryotic protein-coding genes are present in only two copies per genome, occurring in the same position on **homologous** chromosomes, one of which is received from each parent. If the two copies differ slightly they are called alleles. Changes in nucleotide sequences are termed mutations or polymorphisms, depending on their effect.

Some genes code not for protein but for RNA molecules that have their own functions within the cell. These include the transfer RNAs, ribosomal RNAs, and a variety of other smaller RNAs with roles in the nucleus. RNA-coding genes are usually present in multiple copies per eukaryotic genome.

Gene Expression

Expression of protein-coding genes begins with the process of **transcription**. During transcription, the helix is unwound, and an **enzyme** (RNA polymerase) binds to the DNA. It then moves along the DNA, and beginning slightly “downstream” at the so-called initiation site, it copies one of the strands to form a molecule of RNA. Transcription ceases when the polymerase reaches a special DNA sequence called the termination site, usually a region high in G-Cs followed by A-Ts.

amino acids building blocks of protein

base pairs two nucleotides (either DNA or RNA) linked by weak bonds

prokaryotes single-celled organisms without a nucleus

homologous carrying similar genes

transcription messenger RNA formation from a DNA sequence

enzyme a protein that controls a reaction in a cell

In prokaryotes, this RNA product is ready to use for protein synthesis, and is called messenger RNA (mRNA). After the mRNA of a gene is formed, it is used by the cell in protein synthesis (**translation**) at the ribosomes.

Thus, the prokaryotic gene consists of an RNA binding site (called the “promoter”), a transcription initiation site, the coding region, and a termination signal. The initiation site should not be confused with the start signal for protein synthesis, nor the termination site with the stop signal in protein synthesis. Each of the translation signals is within the coding region, or “open reading frame,” of the gene.

Eukaryotic Genes

In eukaryotic cells, genes are more complex. It was discovered in 1977 that eukaryotic genes are functionally separated into coding segments called exons, which are interrupted by noncoding sequences of DNA called introns. The entire region between the initiation and termination sites is transcribed, including the introns, to form the primary transcript. This must then be processed by special enzymes that cut out the introns and splice together the exons to form an mRNA. The mRNA is then exported from the nucleus for translation.

The existence of introns allows for the creation of multiple proteins from one gene, by the use or exclusion of different exons. Such alternative splicing gives rise to protein “isoforms,” highly similar but slightly different proteins, with functions that vary as well. Isoforms are typically tissue-specific. For example, the muscle enzyme creatine kinase exists in one form in the heart, and another form in the skeletal muscles (such as the biceps), which have different ends formed through use of different exons. Even though it codes for two or more proteins, most scientists call such a DNA sequence a single gene.

Eukaryotic genes also contain a sequence close to the termination site called the polyadenylation signal. After transcription, this sequence prompts a special enzyme, called poly-A polymerase, to cut the RNA chain and begin adding multiple adenine nucleotides, as many as 250, to the primary transcript. This poly-A tail helps transport the RNA out of the nucleus, stabilizes it in the **cytoplasm**, and promotes efficient transcription at the ribosome.

Thus, the eukaryotic gene consists of an RNA binding site (promoter), a transcription initiation site, the coding region including exons and introns, the polyadenylation signal, and a termination site.

Genes for RNAs are transcribed in the same way, but the RNA formed is not translated into protein. Details vary among different types, but most RNA-coding genes do not contain introns. Transcripts of the ribosomal RNA genes must be cut apart to form a number of smaller functional RNA molecules.

Controlling Gene Expression

The complexity of any living cell is due to the well-orchestrated interactions of its proteins. Just as an orchestra cannot have every instrument play at once, a cell cannot have all its proteins function at once. One method of regulating protein function is to control when the protein is made, which

translation synthesis of protein using mRNA code

cytoplasm the material in a cell, excluding the nucleus



transcription factors
proteins that increase
the rate of gene trans-
cription

is to say when the gene is expressed. Prokaryotic genes are usually controlled by operon systems, relatively simple systems that tie expression directly to metabolic activity in the cell. Eukaryotic genes are controlled by more complex regulatory systems that respond to hormones, growth factors, internal conditions, and many other influences.

To ensure that each gene is expressed when, and only when, it is needed, each eukaryotic gene has several control regions, termed the promoter and enhancer regions. These do not code for amino acids but are critical for proper gene expression. Mutations in these regions often change the rate at which a gene is expressed, or the factors in the cell or the environment to which it responds.

The promoter region is a sequence of 20 to 200 nucleotides “upstream” of the coding region to which the RNA polymerase enzyme binds, permitting it to begin transcribing the DNA. Promoters differ in size and sequence in prokaryotic and eukaryotic genes. Promoters attract RNA polymerase by first binding a variety of other proteins, called **transcription factors**. In some eukaryotic genes, promoter sites also occur within the coding region, allowing alternative transcripts with fewer exons.

Enhancers, also called activation sites, are located either nearby or far away from the promoter. Because DNA is looped and coiled, however, these sites are actually physically close to the gene’s promoter even when distant on the DNA strand. Enhancers are gene-specific, and attract a variety of transcription factors. All of these work together to increase the rate of transcription by increasing the likelihood of RNA polymerase binding. Controlling the availability of these proteins is an important factor in regulating expression of the gene. SEE ALSO CHROMOSOME, EUKARYOTIC; CHROMOSOME, PROKARYOTIC; CRICK, FRANCIS; DNA; EVOLUTION OF GENES; GENE EXPRESSION: OVERVIEW OF CONTROL; GENE FAMILIES; GENETIC CODE; MENDEL, GREGOR; MORGAN, THOMAS HUNT; MUSCULAR DYSTROPHY; MUTATION; NATURE OF THE GENE, HISTORY; NUCLEOTIDE; OPERON; PROTEINS; RNA POLYMERASES; RNA PROCESSING; TRANSCRIPTION; TRANSCRIPTION FACTORS; WATSON, JAMES.

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phenotype observable
characteristics of an
organism

Gene and Environment

Questions of “nature versus nurture” have been asked of most human traits: Is it our genes, inherited from our parents, that make us the way we are, or is it the environment in which we live? A **phenotype** is a trait that can be observed and described in a population. Although some phenotypes may be

totally controlled by genetic or environmental factors, most are influenced by a complex combination of the two. Genes and environmental factors may work independently, or they may interact with one another to cause the phenotype.

Classes of Human Genetic Phenotypes

Human phenotypes are often classified as either simple (or Mendelian) or complex. A “simple” or Mendelian phenotype is one that demonstrates a recognizable inheritance pattern (such as autosomal dominant or recessive, or X-linked). A Mendelian phenotype is caused by a particular genetic variant, or **allele**, of a gene. The expression of Mendelian phenotypes may vary by age, but, in general, the effect of a single gene is sufficient to cause the phenotype.

allele a particular form of a gene

In contrast, “complex” phenotypes do not adhere to simple Mendelian laws, and they are influenced by several factors (either genetic or environmental) acting independently or together. In complex phenotypes, alleles of particular genes increase the probability that the phenotype will develop, but do not determine with certainty whether a person will have the phenotype. They are neither necessary nor sufficient to cause the phenotype. Genes that act in this fashion are called susceptibility genes. The complex interaction of susceptibility genes with other genetic and environmental risk factors determines whether or not a person will develop a complex phenotype.

Gene-Environment Interaction in Phenylketonuria

Phenylketonuria (PKU) is a classic example of gene-environment interaction. PKU was originally described as an autosomal recessive metabolic disease, in which people with two defective copies of the phenylalanine hydroxylase gene are unable to convert phenylalanine into tyrosine. This inability leads to an accumulation of phenylalanine in the blood, causing problems with nerve and brain development that result in mental retardation.

The treatment of PKU by removing foods containing phenylalanine from the diet (and thus reducing the accumulation of phenylalanine) demonstrated that mutations in the phenylalanine hydroxylase gene cause mental retardation only in the presence of dietary phenylalanine. Since phenylalanine is very common in the diet, this gene-environment interaction was not detected at first. PKU serves as an illustration that phenotypes that are apparently Mendelian in nature may have complex interactions with other genes and with the environment. Removing the exposure to dietary phenylalanine prevents mental retardation, and phenylalanine does not cause mental retardation in the absence of **mutations** in the phenylalanine hydroxylase gene. Therefore, both factors are needed to cause mental retardation due to PKU.

mutations changes in DNA sequences

The identification of the gene-environment interaction in PKU has led to the effective treatment of this genetic disorder. Individuals who carry mutations in the phenylalanine hydroxylase gene, if placed on a low-phenylalanine diet, generally do not develop the symptoms of PKU. To identify individuals at risk of PKU, newborns are screened for elevated phenylalanine levels in the blood. Those infants with positive screening tests are then evaluated further. Those with PKU (about 1 in 10,000 live



epidemiologists person who studies the incidence and spread of diseases in a population

phenotype observable characteristics of an organism

births) are then placed on low-phenylalanine diets to prevent the development of mental retardation.

Methods for Identifying Gene-Environment Interactions

As the example of PKU demonstrates, studies that attempt to identify factors important in determining human phenotypes must simultaneously examine multiple genetic and environmental factors. It is generally not possible to experiment directly on humans to observe the effects of a gene or environmental factor on the expression of a phenotype. Human genetic studies are generally observational studies, where the researcher is limited to observing the combinations of exposures that naturally occur in the population. Genetic **epidemiologists** must apply statistical methods to these observational data to evaluate how genes and the environment affect the development of a **phenotype**.

The simultaneous analysis of genetic and environmental factors allows the identification of environmental factors that interact with each other. Researchers use statistical models to compare the joint effects of genetic and environmental factors in people with the phenotype and people without the phenotype. Such “case-control studies” are commonly used to examine the relationship between disease phenotypes and both genetic and nongenetic risk factors.

The strength of the association between a risk factor and the disease is described by the “relative risk,” which is the probability of having the phenotype if exposed to the risk factor divided by the probability of having the phenotype if not exposed to the risk factor. A relative risk greater than 1 suggests that the risk factor increases the probability of developing the phenotype, whereas a relative risk less than 1 suggests the risk factor decreases the probability of the phenotype. An association between a risk factor and a phenotype exists if the relative risk is significantly different from 1.

An example of a risk factor and relative risk can be seen in Alzheimer’s disease (AD). One gene that influences the risk of AD is the *APOE* gene. Three common alleles are known: ϵ -2, ϵ -3, and ϵ -4. Caucasian Americans with one or more copies of ϵ -4 are two and one-half times more likely to develop AD than are people with two ϵ -3 alleles. Interestingly, the ϵ -4 allele is not as strong of a risk factor for AD among African Americans or Hispanics, who nonetheless have higher risks of developing AD than Caucasians.

Patterns of Gene-Environment Interactions

There are many potential patterns of interaction that could exist between genetic and environmental factors for a phenotype. Several plausible statistical models of gene-environment interaction have been described. Phenotype expression can either be:

- (1) Increased only in the presence of both the susceptibility genotype and the environmental factor;
- (2) Increased by the environmental factor alone but not by the genotype alone;
- (3) Increased by the genotype alone but not by the environmental factor alone;